

$$i = \frac{dq}{dt} \quad v = \frac{dw}{dq} \quad p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = iv$$

$$i = C \frac{dv}{dt} \quad v = L \frac{di}{dt}$$

$$v = iR \quad R_1 || R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$p = Cv \frac{dv}{dt} \quad p = Li \frac{di}{dt}$$

$$p = iv = i^2 R = \frac{v^2}{R}$$

$$w = \frac{1}{2} Li^2 \quad w = \frac{1}{2} Cv^2$$

$$R = \frac{v}{i}$$

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)] e^{-(t-T_0)/\tau}$$

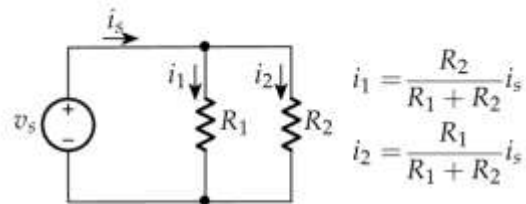
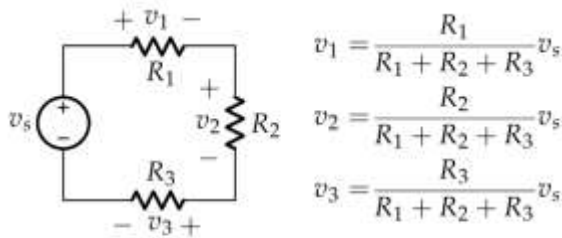
$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)] e^{-(t-T_0)/\tau}$$

(Switch action at $t = T_0$)

ODE: $\frac{dx}{dt} + ax = b$, General solution: $x(t) = x(0)e^{-at} + \frac{b}{a}(1 - e^{-at})$

ODE: $\frac{dx}{dt} + ax = 0$, General solution: $x(t) = x(0)e^{-at}$

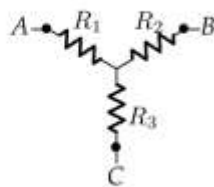
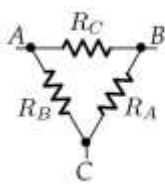
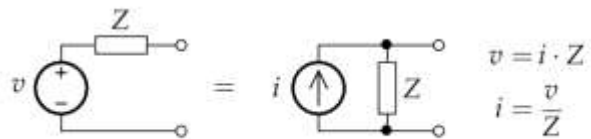
Note that: $\frac{b}{a} = x(\infty)$, $a = \frac{1}{\tau}$



$$v_B = \underbrace{V_B}_{\text{DC}} + \underbrace{v_b}_{\text{AC}}$$

$\bar{V}_b \rightarrow$ Phasor notation

$$\text{CMRR} = 20 \log_{10} \frac{|A_d|}{|A_{cm}|}$$



$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$

Table 6-2: General solution for second-order circuits for $t \geq 0$.

$x(t)$ = unknown variable (voltage or current)	
Differential equation:	$x'' + ax' + bx = c$
Initial conditions:	$x(0)$ and $x'(0)$
Final condition:	$x(\infty) = \frac{c}{b}$
$\alpha = \frac{a}{2}$	$\omega_0 = \sqrt{b}$
Overdamped Response $\alpha > \omega_0$	
$x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)] u(t)$	
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$
$A_1 = \frac{x'(0) - s_2[x(0) - x(\infty)]}{s_1 - s_2}$	$A_2 = -\left[\frac{x'(0) - s_1[x(0) - x(\infty)]}{s_1 - s_2} \right]$
Critically Damped $\alpha = \omega_0$	
$x(t) = [(B_1 + B_2 t) e^{-\alpha t} + x(\infty)] u(t)$	
$B_1 = x(0) - x(\infty)$	$B_2 = x'(0) + \alpha[x(0) - x(\infty)]$
Underdamped $\alpha < \omega_0$	
$x(t) = [(D_1 \cos \omega_d t + D_2 \sin \omega_d t) e^{-\alpha t} + x(\infty)] u(t)$	
$D_1 = x(0) - x(\infty)$	$D_2 = \frac{x'(0) + \alpha[x(0) - x(\infty)]}{\omega_d}$
$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$	
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \frac{1}{\sqrt{LC}}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$$v(t) = V_m \cos(\omega t + \theta_v); i(t) = I_m \cos(\omega t + \theta_i);$$

$$p(t) = \frac{V_m I_m}{2} \cos \theta + \frac{V_m I_m}{2} \cos \theta \cos(2\omega t) -$$

$$\frac{V_m I_m}{2} \sin \theta \sin(2\omega t);$$

$$P = p_{avg} = \frac{V_m I_m}{2} \cos \theta; \quad Q = \frac{V_m I_m}{2} \sin \theta; \quad \theta = \theta_v -$$

$$\theta_i; \quad Z = \frac{V}{I} = \frac{V_m}{I_m} \angle \theta;$$

$$\cos \theta: \text{power factor, p.f.}; \quad \sin \theta: \text{reactive factor, r.f.};$$

$$\theta: \text{power factor angle};$$

$$S = P + jQ; \text{ Units: } S: [\text{VA}]; P: [\text{W}]; Q: [\text{VAR}];$$

$$S = \frac{V_m I_m^*}{2} = V_{rms} I_{rms}^* ;$$

$$P = |I_{rms}|^2 R; Q = |I_{rms}|^2 X; Z = R + jX;$$

$$S = \frac{|V_{rms}|^2}{Z^*} ;$$

